Mathematical Notation for Statistics

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Contents

1 Notation Conventions

We use the following conventions throughout:

- Scalars: x, μ, σ^2 (lowercase italic)
- Vectors: $\mathbf{x}$, $\boldsymbol{\beta}$ (bold lowercase)
- Matrices: $\mathbf{X}$, $\boldsymbol{\Sigma}$ (bold uppercase)
- Sets: $\mathcal{X}$, $\mathcal{F}$ (calligraphic)
- Random variables: X, Y (uppercase italic)

2 Probability and Expectation

Let X and Y be random variables on a probability space $(\Omega, \mathcal{F}, \Pr)$.

$$\mathbb{E}[X] = \int x\, f(x)\, dx \tag{1}$$

$$\operatorname{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] \tag{2}$$

$$\operatorname{Cov}(X, Y) = \mathbb{E}\left[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\right] \tag{3}$$

We write $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ to mean that the X_i are independent and identically distributed.

3 Statistical Models

3.1 Linear Model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n) \tag{4}$$

The OLS estimator minimizes $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \tag{5}$$

3.2 Logistic Regression

For a binary outcome $Y_i \in \{0, 1\}$:

$$\log \frac{p_i}{1 - p_i} = \mathbf{x}_i^\top \boldsymbol{\beta}, \qquad p_i = \Pr(Y_i = 1 \mid \mathbf{x}_i) \tag{6}$$

3.3 Cox Proportional Hazards

$$h(t \mid \mathbf{x}) = h_0(t) \exp\bigl(\mathbf{x}^\top \boldsymbol{\beta}\bigr) \tag{7}$$

where $h_0(t)$ is the unspecified baseline hazard.

4 Likelihood and Information

The log-likelihood is:

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n} \log f(y_i \mid \mathbf{x}_i, \boldsymbol{\theta}) \tag{8}$$

The Fisher information matrix is:

$$\mathcal{I}(\boldsymbol{\theta}) = -\mathbb{E}\left[\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}^\top}\right] \tag{9}$$

5 Theorem Environments

Definition 5.1 (Unbiased estimator)**.** An estimator $\hat{\theta}$ is *unbiased* if $\mathbb{E}[\hat{\theta}] = \theta$ for all θ.

Theorem 5.1 (Cramér-Rao lower bound)**.** *For any unbiased estimator $\hat{\theta}$ of a scalar parameter θ,*

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{\mathcal{I}(\theta)} \tag{10}$$

where $\mathcal{I}(\theta)$ is the Fisher information.

Proof. Apply the Cauchy-Schwarz inequality to $\mathrm{Cov}\Bigl(\hat{\theta}, \partial_\theta \log f(X; \theta)\Bigr)$. The score has mean zero and variance $\mathcal{I}(\theta)$, which gives the result. □

Corollary 5.2. *The OLS estimator in* (**??**) *is the minimum-variance unbiased linear estimator (Gauss-Markov theorem).*